Vector-Valued Functions

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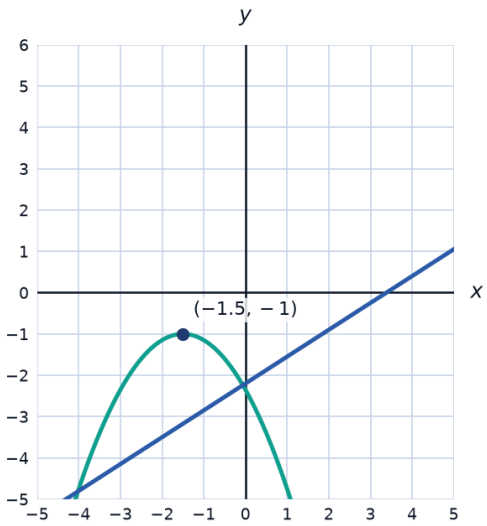
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Position, velocity, and acceleration vectors

A particle's position is given by $\vec{r}(t) = \langle t^2, t^3 \rangle$. Find the velocity vector $\vec{v}(t)$ and acceleration vector $\vec{a}(t)$.

2 [Foundational] • Target: Position, velocity, and acceleration vectors

For the same particle, find the speed at $t = 2$.



3 [Foundational] • Target: Integrating vector-valued functions

Given $\vec{v}(t) = \langle 3t^2, 2t \rangle$ and initial position $\vec{r}(0) = \langle 1, 4 \rangle$, find $\vec{r}(t)$.

4 [Foundational] • Target: Total distance traveled

A particle has velocity $\vec{v}(t) = \langle \cos t, \sin t \rangle$ for $t \in [0, \pi]$. Find the total distance traveled.

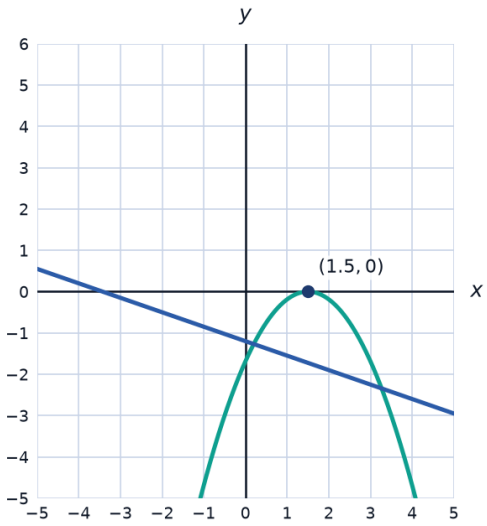
5 [Foundational] • Target: Vectors and parametric motion combined

A particle moves along $x(t) = e^t$, $y(t) = e^{-t}$. Find the velocity vector and speed at $t = 0$.

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6 [Foundational] • Target: Vectors and parametric motion combined

For the particle above, find the magnitude of the acceleration vector at $t = 0$.



7 [Foundational] • Target: Unit tangent vectors

For $\vec{r}(t) = \langle t, t^2 \rangle$, find the unit tangent vector $\vec{T}(t)$ at $t = 1$.

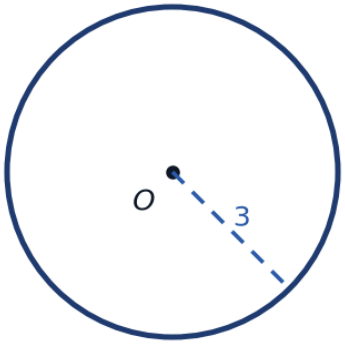
Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Unit tangent vectors

Explain why $\vec{T}(t) = \frac{\vec{v}(t)}{|\vec{v}(t)|}$ always has magnitude 1.

9 [Intermediate] • Target: Motion on a circle and centripetal acceleration

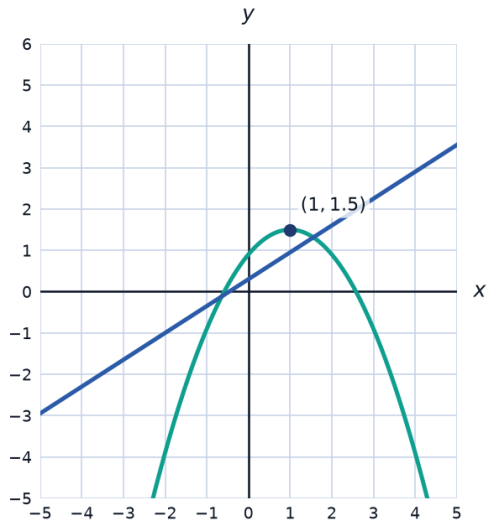
A particle moves along $\vec{r}(t) = \langle 3\cos t, 3\sin t \rangle$. Find $\vec{v}(t)$, $\vec{a}(t)$, and verify that $\vec{a}(t)$ points toward the origin (is a scalar multiple of $-\vec{r}(t)$).



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10 [Intermediate] • Target: Motion on a circle and centripetal acceleration

For the same particle, verify that $\vec{v}(t) \cdot \vec{r}(t) = 0$ for all t, and interpret this geometrically.



11 [Intermediate] • Target: Displacement vs. distance for vector motion

A particle has position $\vec{r}(t) = \langle t^2, t \rangle$ for $t \in [0, 2]$. Find the displacement vector from $t = 0$ to $t = 2$.

12 [Intermediate] • Target: Displacement vs. distance for vector motion

Set up (but do not evaluate) the integral for the total distance traveled by the particle $\vec{r}(t) = \langle t^2, t \rangle$ over $t \in [0, 2]$.

13 [Intermediate] • Target: Vector operations review

Find the dot product $\vec{u} \cdot \vec{w}$ for $\vec{u} = \langle 3, -2 \rangle$ and $\vec{w} = \langle -1, 4 \rangle$, and state whether the vectors are perpendicular.

14 [Intermediate] • Target: Curvature-related speed problems

A particle moves along $\vec{r}(t) = \langle t^3, t^2 \rangle$. Find the times t when the particle's speed is exactly $\sqrt{13}$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

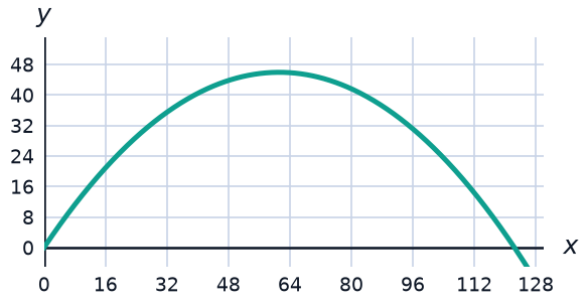
15 [Challenging] • Target: Curvature-related speed problems • Cross-Lesson: 6.01 Power Series and Interval of Convergence

For the particle $\vec{r}(t) = \langle t, \ln t \rangle$ $(t > 0)$, find the velocity vector and speed at $t = 1$.

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16 [Challenging] • Target: Projectile motion as a vector-valued function • Cross-Lesson: 6.02 Taylor and Maclaurin Series

A projectile is launched with position $\vec{r}(t) = \langle 20t,\ 30t - 4.9t^2 \rangle$ (meters, seconds). Find the velocity vector $\vec{v}(t)$ and the time when the projectile reaches maximum height (where the vertical velocity component is 0).



17 [Challenging] • Target: Projectile motion as a vector-valued function • Cross-Lesson: 1.01 Integration by Parts

For the same projectile, find the speed at $t = 0$ (launch) and interpret its two components.

18 [Challenging] • Target: Finding when a particle is at rest • Cross-Lesson: 1.02 Partial Fractions and Improper Integrals

A particle moves along $\vec{r}(t) = \langle t^3 - 3t,\ t^2 - 4 \rangle$. Find all times t when the particle is momentarily at rest (both velocity components equal 0 simultaneously).

19 [Challenging] • Target: Finding when a particle is at rest • Cross-Lesson: 2.01 Euler's Method and Logistic Growth

For the same particle, find the times when only the horizontal velocity component is 0.

20 [Challenging] • Target: Vector-valued functions and arc length • Cross-Lesson: 3.01 Parametric Equations

State the arc length formula for a vector-valued function $\vec{r}(t) = \langle x(t), y(t) \rangle$ on $[a, b]$, and explain how it relates to the speed $|\vec{v}(t)|$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Position, velocity, and acceleration vectors

A particle moves along $\vec{r}(t) = \langle 5t^2,\ 12t^2 \rangle$. Find the magnitude of the (constant) acceleration vector $|\vec{a}(t)|$.

a. 26

b. 28

c. 34

d. 13

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22 [Foundational] • Target: Integrating vector-valued functions

A particle moves along $\vec{r}(t) = \langle t^3 - 12t, t^2 + 3 \rangle$. Find the positive time t when the horizontal velocity component is 0.

a. -2

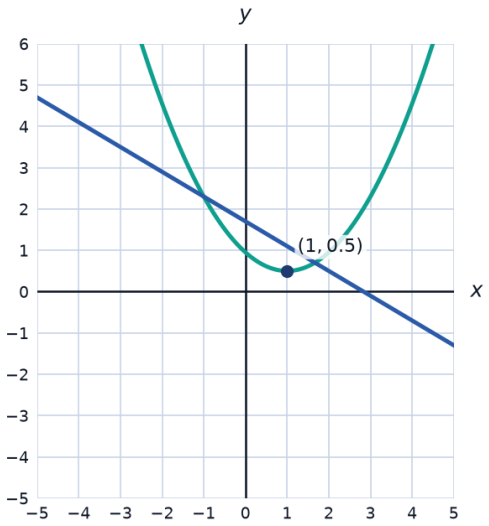
b. 6

c. 2

d. 3

23 [Foundational] • Target: Total distance traveled

A particle moves along $\vec{r}(t) = \langle 7t, 48t - 4t^2 \rangle$. Find the time t when the vertical velocity component is 0 (the time of maximum height).



a. 7

b. 6

c. 12

d. 5

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24 [Foundational] • Target: Vectors and parametric motion combined

Consider the function $f(x) = e^x \sin(x)$. A particle's position on a number line is given by $s(t) = f(t)$ for $t \geq 0$. The particle is momentarily at rest when its velocity is zero. The acceleration of the particle at these moments is given by $a(t) = f''(t)$. Find the smallest positive value of t at which the particle is momentarily at rest, and then determine the acceleration of the particle at that exact moment.

a. Rest at $t = \frac{3\pi}{4}$, acceleration $a = -\sqrt{2}e^{\frac{3\pi}{4}}$

b. Rest at $t = \frac{\pi}{2}$, acceleration $a = e^{\frac{\pi}{2}}$

c. Rest at $t = \frac{3\pi}{4}$, acceleration $a = -\frac{\sqrt{2}}{2}e^{\frac{3\pi}{4}}$

d. Rest at $t = \pi$, acceleration $a = 0$

25 [Foundational] • Target: Unit tangent vectors

A particle moves along the x-axis. Its velocity is given by the function $v(t) = \sqrt{t^2 + 1}$ for $t \geq 0$. The position of the particle at time $t = 0$ is $x(0) = 5$. Consider the function $P(x)$ defined as the total displacement of the particle from $t = 0$ to $t = x$. That is, $P(x) = x(x) - x(0)$. Which of the following statements about $P(x)$ is true for $x > 0$?

a. $P(x)$ is always positive and increasing.

b. $P(x)$ is always negative and decreasing.

c. $P(x)$ is always positive and decreasing.

d. $P(x)$ is always negative and increasing.

26 [Foundational] • Target: Motion on a circle and centripetal acceleration

Unit tangent vector $\mathbf{T}(t) = \mathbf{r}'(t)/|\mathbf{r}'(t)|$. It has magnitude:

a. 1

b. 0

c. $|r'|$

d. t

27 [Foundational] • Target: Displacement vs. distance for vector motion

For a circle of radius R, curvature $\kappa =$

a. R

b. $2R$

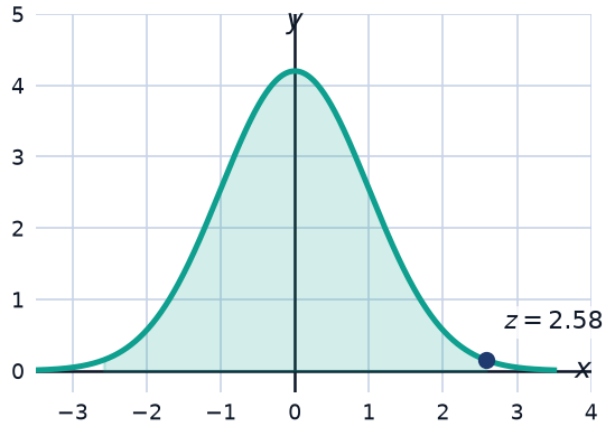
c. $1/R^2$

d. $1/R$

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28 [Foundational] • Target: Vector operations review

Tangent plane at $(0, 0, 1)$ on unit sphere: $z = 1$. Normal direction:



a. $(0, 1, 0)$

b. $(1, 0, 0)$

c. $(0, 0, 1)$

d. $(1, 1, 1)$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Curvature-related speed problems

A particle's position in the complex plane is given by $z(t) = (2t^2 - 3t + 1) + i(t^3 - 6t^2 + 11t - 6)$, where t is time in seconds and $t \geq 0$. The particle starts at $t = 0$ and its path is traced by the complex number $z(t)$. The particle is said to be 'at rest' when its velocity, represented by the complex number $v(t) = z'(t)$, is zero. Determine the times $t > 0$ when the particle is at rest.

a. $t = 1$ and $t = 3$

b. $t = 1$ and $t = 2$

c. $t = 2$ and $t = 3$

d. $t = 1, t = 2,$ and $t = 3$

30 [Intermediate] • Target: Projectile motion as a vector-valued function

A particle's velocity is given by $v(t) = 3t^2 - 12t + 9$ for $t \geq 0$. The position of the particle at time $t = 0$ is $s(0) = 5$. \\ Consider a second particle whose position $p(t)$ is described by the integral $p(t) = \int_0^t v(u)\,du$. \\ Find the net displacement of the first particle from $t = 1$ to $t = 3$.

a. -4

b. 4

c. 8

d. -8

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31 [Intermediate] • Target: Finding when a particle is at rest

The velocity of a particle moving along a straight line is given by the function $v(t) = t^2 - 6t + 8$ m/s. What is the instantaneous acceleration of the particle at time $t = 3$ seconds?

a. -1 m/s^2

b. 0 m/s^2

c. 1 m/s^2

d. -2 m/s^2

32 [Intermediate] • Target: Vector-valued functions and arc length

A particle's velocity is given by the function $v(t) = 5$ for $0 \le t \le 4$. Find the total displacement of the particle over this time interval.

a. 10

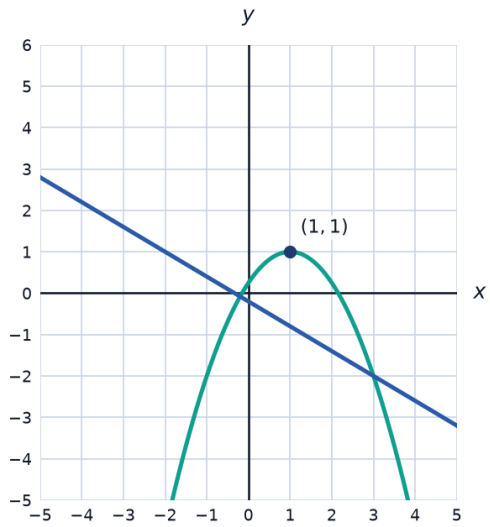
b. 15

c. 20

d. 25

33 [Intermediate] • Target: Position, velocity, and acceleration vectors

Distance between skew lines $L_1 : \mathbf{r} = \mathbf{a} + s\mathbf{d}_1$ and $L_2 : \mathbf{r} = \mathbf{b} + t\mathbf{d}_2$:



a. $|\mathbf{b} - \mathbf{a}|$

b. $|(\mathbf{b} - \mathbf{a}) \cdot (\mathbf{d}_1 \times \mathbf{d}_2)|/|\mathbf{d}_1 \times \mathbf{d}_2|$

c. $|\mathbf{d}_1 - \mathbf{d}_2|$

d. 0

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34 [Intermediate] • Target: Integrating vector-valued functions

Parametric surface $\mathbf{r}(u, v) = (u\cos v, u\sin v, u^2)$. At $(u, v) = (1, 0)$:

a. tangent plane is $z = x^2$

b. tangent plane is $z = 2x - 1$

c. normal is $(0, 0, 1)$

d. surface is a sphere

35 [Intermediate] • Target: Total distance traveled

Find the curvature of $\mathbf{r}(t) = (t, t^2, 0)$ at $t = 0$:

a. 2

b. 1

c. 0

d. 1/2

36 [Intermediate] • Target: Vectors and parametric motion combined

For $\mathbf{r}(t) = \langle 2t, 3t^2, t\rangle$, find the normal component of acceleration at $t = 1$.

a. $\frac{6\sqrt{205}}{41}$

b. $\frac{36\sqrt{41}}{41}$

c. 6

d. 36

37 [Intermediate] • Target: Unit tangent vectors

For $\mathbf{r}(t) = \langle 2t, t^2, t\rangle$, find the tangential component of acceleration at $t = 2$.

a. $\frac{2\sqrt{105}}{21}$

b. $\frac{8\sqrt{21}}{21}$

c. 2

d. 8

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Motion on a circle and centripetal acceleration • Cross-Lesson: 1.02 Partial Fractions and Improper Integrals

The torsion τ of a space curve measures how fast the curve twists out of its osculating plane. For a helix $\mathbf{r} = (a\cos t, a\sin t, bt)$, as $b \to 0$ (curve flattens to a circle), $\tau \to$

a. $1/a$

b. 0

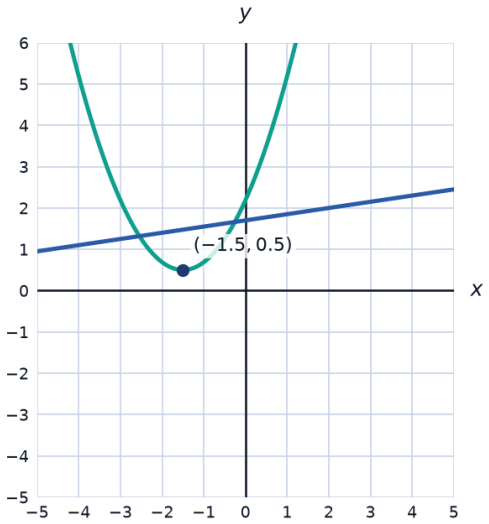
c. ∞

d. b

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39 [Challenging] • Target: Displacement vs. distance for vector motion • Cross-Lesson: 2.01 Euler's Method and Logistic Growth

Distance from point $P = (1, 2, 3)$ to plane $2x + y - 2z = 5$:



a. $\frac{7}{3}$

b. $\frac{2}{3}$

c. 1

d. 3

40 [Challenging] • Target: Vector operations review • Cross-Lesson: 3.01 Parametric Equations

Divergence theorem: $\oiint_S \mathbf{F} \cdot d\mathbf{S} = \iiint_V (abla \cdot \mathbf{F})\, dV$. For $\mathbf{F} = \mathbf{r} = (x, y, z)$, $abla \cdot \mathbf{F} = 3$: flux through unit sphere =

a. 3

b. $3V = 4\pi$

c. 4π

d. 12π

41 [Challenging] • Target: Curvature-related speed problems • Cross-Lesson: 4.01 Polar Curves and Derivatives

Position vector of midpoint of segment from $A(1, 2, 3)$ to $B(5, 6, 7)$:

a. $(6, 8, 10)$

b. $(2, 2, 2)$

c. $(3, 4, 5)$

d. $(4, 4, 4)$

42 [Challenging] • Target: Projectile motion as a vector-valued function • Cross-Lesson: 4.02 Area in Polar Coordinates

Curvature of helix $\mathbf{r} = (a\cos t, a\sin t, bt)$:

a. ab

b. $b/(a^2 + b^2)$

c. $1/a$

d. $a/(a^2 + b^2)$

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43 [Challenging] • Target: Finding when a particle is at rest • Cross-Lesson: 5.01 Sequences and Basic Series Convergence Tests

Potential function for $\mathbf{F} = (2xy, x^2 + 3y^2, 0)$:

a. $f = x^2y + y^3$

b. $f = x^2y^2$

c. $f = xy^2$

d. $f = x^2 + y^2$

44 [Challenging] • Target: Vector-valued functions and arc length • Cross-Lesson: 5.02 Ratio, Root, and Alternating Series Tests

Find the distance from $(2,\ -2,\ 0)$ to the plane $-3y + z = -4$.

a. 10

b. $\frac{\sqrt{10}}{5}$

c. 1

d. $\sqrt{10}$

45 [Challenging] • Target: Position, velocity, and acceleration vectors • Cross-Lesson: 6.01 Power Series and Interval of Convergence

Stokes' theorem for $\mathbf{F} = (y,\ -x, 0)$ over unit disk $z = 0$: $\oint \mathbf{F} \cdot d\mathbf{r} =$

a. 0

b. 2π

c. -2π

d. π

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